

LESSON 5-8 • SESSION 1 • PREREQUISITE PRACTICE A • SKILL 1 OF 2

Multiply Equal Face Areas

Builds on Grade 4 • 4.NBT.B.5

TODAY'S LESSON

You will find the 4 equal triangles of a square pyramid, like $4 \times (\frac{1}{2} \times 8 \times 11)$ for the gift topper.

YOU NEED TO

Multiply one face by 4, like 4×24 . **You will use it in Practice Form A, problems 3, 4, and 7.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1 Find** the area of one face.
- 2 Double** it.
- 3 Double** again. That is 4 times the face.

WATCH ME*the steps in action***Four equal faces are each 24 in². Find their total area.****Step 1** One face: 24 in².**Step 2** Double 24 is 48.**Step 3** Double 48 is 96.**Answer 96 in²****TRY IT WITH HELP***fill in each step*

- 1.** Four equal faces are each 35 cm². Find their total area.

Step 1 One face: _____ cm²**Step 2** Double 35 is _____**Step 3** Double 70 is _____ cm²

- 2.** Four equal faces are each 18 m². Find their total area.

Step 1 One face: _____ m²**Step 2** Double 18 is _____**Step 3** Double 36 is _____ m²**YOUR TURN***use the steps on your own*

- 3.** Multiply.

a. $4 \times 12 =$ _____

b. $4 \times 25 =$ _____

c. $4 \times 45 =$ _____

d. $4 \times 16 =$ _____

e. $4 \times 60 =$ _____

f. $4 \times 75 =$ _____

Area of a Square Base

Builds on Grade 4 • 4.MD.A.3

TODAY'S LESSONYou will find a pyramid's square base, like $16\text{ cm} \times 16\text{ cm}$.**YOU NEED TO**Find a square's area: side $\times$ side, like 19×19 . **You will use it in Practice Form A, problems 2, 4, and 7.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Break** the side into tens and ones.
- 2 Multiply** each part by the side.
- 3 Add** the parts. Write square units.

WATCH ME*the steps in action***A square base is 19 cm on each side. Find its area.****Step 1** $19 = 10 + 9$.**Step 2** $10 \times 19 = 190$ and $9 \times 19 = 171$.**Step 3** $190 + 171 = 361$, so 361 cm^2 .**Answer** **361 cm^2** **TRY IT WITH HELP***fill in each step***4.** A square base is 13 in on each side. Find its area.**Step 1** $13 = 10 + \underline{\hspace{2cm}}$ **Step 2** $10 \times 13 = \underline{\hspace{2cm}}$ and $3 \times 13 = \underline{\hspace{2cm}}$ **Step 3** $130 + 39 = \underline{\hspace{2cm}}\text{ in}^2$ **5.** A square base is 15 ft on each side. Find its area.**Step 1** $15 = 10 + \underline{\hspace{2cm}}$ **Step 2** $10 \times 15 = \underline{\hspace{2cm}}$ and $5 \times 15 = \underline{\hspace{2cm}}$ **Step 3** $150 + 75 = \underline{\hspace{2cm}}\text{ ft}^2$ **YOUR TURN***use the steps on your own***6.** Find each square's area.**a.** $12 \times 12 = \underline{\hspace{2cm}}$ **b.** $14 \times 14 = \underline{\hspace{2cm}}$ **c.** $17 \times 17 = \underline{\hspace{2cm}}$ **d.** $18 \times 18 = \underline{\hspace{2cm}}$ **e.** $21 \times 21 = \underline{\hspace{2cm}}$ **f.** $25 \times 25 = \underline{\hspace{2cm}}$ **READY CHECK***a problem like the one in today's lesson***7.** A square pyramid has a 10 cm base. Each triangle is 35 cm^2 . What is its surface area?**A** 35 cm^2 **B** 135 cm^2 **C** 140 cm^2 **D** 240 cm^2 **CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #